SYNTHETIC EXAM — TEST DATA ONLY

Q1. A 12 V source is connected across a 4 ohm resistor.

What is the current? Show your calculation.

Student work:

V = I R

I = V / R

I = 12 / 4 = 4 A

(The result above intentionally contains an arithmetic mistake.)

Q2. Calculate the impedance of a series R-L circuit.

Assume R = 4 ohm and inductive reactance is 3 ohm.

Student work:

Z = R + j XL

Z = 4 + j3 ohm

|Z| = sqrt(4^2 + 3^2) = 5 ohm

Angle = atan(3/4) = 36.9 degrees